

# 实验四 探究人类基因组第 21 号染色体

刘沛雨 2100012289 信息科学技术学院

## 练习 1

通过 ssh 连接到教学集群（计 1 分）

```
[(base) peiyuliu@localhost Week5 % ssh -i ./yy.txt -p 16622 teach80_pkuhpc@162.105.160.16
Activate the web console with: systemctl enable --now cockpit.socket
```

```
Last login: Tue Mar 18 13:24:43 2025 from 11.250.1.1
:)Welcome to PPKUHPG!
+++++ System Data :+++++
+ Hostname = login66
+ Uptime = 13:26:32 up 19 days, 22:17
+ Free Memory = 114 G
+++++ User Data :+++++
+ Username = teach80_pkuhpc
+ Sessions = 1 of Unlimited MAX
+ Processes = 918 of 512429 MAX
any problem, you can send mail to:
fangt@cls.pku.edu.cn zws@cls.pku.edu.cn chenfbj@pku.edu.cn
+++++ filesystem quota: Please wait 2 seconds +++++
```

```
+-----+
|files|show|used|free|quota|warning|
+-----+
|home|blocks|0.00(TB)|0.88(TB)|0.88(TB)|none|
|lustre1|blocks|0.00(TB)|1.12(TB)|1.12(TB)|none|
|-----|inodes|0.0(K)|10000.0(K)|10000.0(K)|none|
|gpfs|blocks|0.00(TB)|0.88(TB)|0.88(TB)|none|
|-----|inodes|0.0(K)|9408.7(K)|9408.7(K)|none|
+-----+
```

提交一个脚本到计算节点

```
[teach80_pkuhpc@login66 ~]# cat ./test.sh
#!/bin/bash

echo "Hello, World!"
[teach80_pkuhpc@login66 ~]# chmod +x ./test.sh
[teach80_pkuhpc@login66 ~]# pkurun-cnlong 1 1 /home/teach80_pkuhpc/test.sh
Submitted batch job 676
[teach80_pkuhpc@login66 ~]# ls -lrt
total 9
lrwxrwxrwx 1 root root 29 Aug 30 2024 wwwroot -> /gpfs1/wwwroot/teach80_pkuhpc
lrwxrwxrwx 1 root root 28 Aug 30 2024 lustre1 -> /lustre1/user/teach80_pkuhpc
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 223 Mar 18 13:29 job.srp132907
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 0 Mar 18 13:29 '$'\342\200\234'\132907_617.out'
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 123 Mar 18 13:29 '$'\342\200\234'\132907_617.err'
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 222 Mar 18 13:29 job.srp132925
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 0 Mar 18 13:29 '$'\342\200\234'\132925_622.out'
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 123 Mar 18 13:29 '$'\342\200\234'\132925_622.err'
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 226 Mar 18 13:30 job.srp133010
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 0 Mar 18 13:30 sle133010_631.err
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 21 Mar 18 13:30 sle133010_631.out
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 240 Mar 18 13:37 job.srp133714
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 0 Mar 18 13:37 tes133714_656.out
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 136 Mar 18 13:37 tes133714_656.err
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 240 Mar 18 13:37 job.srp133738
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 0 Mar 18 13:37 tes133738_658.err
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 16 Mar 18 13:37 tes133738_658.out
-rwxrwxr-x 1 teach80_pkuhpc teach80_pkuhpc 34 Mar 18 13:38 test.sh
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 240 Mar 18 13:38 job.srp133833
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 0 Mar 18 13:38 tes133833_660.err
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 14 Mar 18 13:38 tes133833_660.out
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 240 Mar 18 13:42 job.srp134205
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 0 Mar 18 13:42 tes134205_676.err
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 14 Mar 18 13:42 tes134205_676.out
[teach80_pkuhpc@login66 ~]# cat ./tes134205_676.out
Hello, World!
[teach80_pkuhpc@login66 ~]#
```

## 练习 2 (1.5 分)

chr21\_stats.sh 文件内容如下:

```
#!/bin/bash

path="/gpfs1/share/class2/chr21.fa"
total_length=$(wc -c $path | awk '{print $1}')
A_counts=$(grep -io 'a' $path | wc -l)
echo "The length of Chromosome21 is $total_length bp."
echo "The counts of Adenine in Chromosome21 is $A_counts."
```

提交脚本到节点运行 (计 0.5 分), 结果如下:

```
[teach80_pkuhpc@login66 peiyuliu]# pkurun-cnlong 1 10 ./chr21_stats.sh
Submitted batch job 1025
[teach80_pkuhpc@login66 peiyuliu]# sq
JOBID PARTITION NAME ST TIME NODES NODELIST(REASON)
[teach80_pkuhpc@login66 peiyuliu]# ls -lrt
total 7
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 226 Mar 18 14:39 job.srp143942
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 0 Mar 18 14:39 chr143942_849.out
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 122 Mar 18 14:39 chr143942_849.err
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 226 Mar 18 14:40 job.srp144020
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 23 Mar 18 14:40 chr144020_852.out
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 99 Mar 18 14:40 chr144020_852.err
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 228 Mar 18 14:40 job.srp144034
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 23 Mar 18 14:40 chr144034_855.out
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 99 Mar 18 14:40 chr144034_855.err
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 228 Mar 18 14:42 job.srp144214
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 0 Mar 18 14:42 chr144214_866.err
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 44 Mar 18 14:42 chr144214_866.out
-rwxrwxr-x 1 teach80_pkuhpc teach80_pkuhpc 249 Mar 18 15:07 chr21_stats.sh
-rwx----- 1 teach80_pkuhpc teach80_pkuhpc 229 Mar 18 15:08 job.srp150814
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 0 Mar 18 15:08 chr150814_1025.err
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 94 Mar 18 15:08 chr150814_1025.out
[teach80_pkuhpc@login66 peiyuliu]# cat ./chr150814_1025.out
The length of Chromosome21 is 47644190 bp.
The counts of Adenine in Chromosome21 is 11820664.
[teach80_pkuhpc@login66 peiyuliu]#
```

21 号染色体的长度为 47644190 bp (计 0.5 分)。

计算得到腺嘌呤占全长的比例约为 24.81% (计 0.5 分)。

```
[teach80_pkuhpc@login66 peiyuliu]# python3
Python 3.6.8 (default, Apr 24 2024, 21:55:04)
[GCC 8.5.0 20210514 (Red Hat 8.5.0-22)] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> 11820664 / 47644190
0.2481029481244198
>>>
```

## 练习 3

首先使用脚本筛选出所有基因以及基因类型。

```
#!/bin/bash

anno_path="/gpfs1/share/class2/annotation.gtf"
awk -F '\t' ' $3=="gene"{print $0}' $anno_path > /home/teach80_pkuhpc/peiyuliu/genes.txt
awk -F '\t' ' $3=="gene"{print $9}' $anno_path | awk -F ' ' '{print $4}' > /home/teach80_pkuhpc/peiyuliu/gene_types.txt
sort /home/teach80_pkuhpc/peiyuliu/gene_types.txt | uniq > uniq_gene_types.txt
```

```

[teach80_pkuhpc@login66 peiyuliu]# wc -l genes.txt
872 genes.txt
[teach80_pkuhpc@login66 peiyuliu]# wc -l gene_types.txt
872 gene_types.txt
[teach80_pkuhpc@login66 peiyuliu]# wc -l uniq_gene_types.txt
15 uniq_gene_types.txt
[teach80_pkuhpc@login66 peiyuliu]# grep -i "protein_coding" ./gene_types.txt | wc -l
232
[teach80_pkuhpc@login66 peiyuliu]# █

```

总的 gene 数量为 872；protein\_coding gene 数量为 232。

有 15 种类型的基因，如下所示：

```

"IG_V_gene";
"TEC";
"lncRNA";
"miRNA";
"misc_RNA";
"processed_pseudogene";
"protein_coding";
"pseudogene";
"rRNA";
"rRNA_pseudogene";
"snRNA";
"snoRNA";
"transcribed_processed_pseudogene";
"transcribed_unprocessed_pseudogene";
"unprocessed_pseudogene";

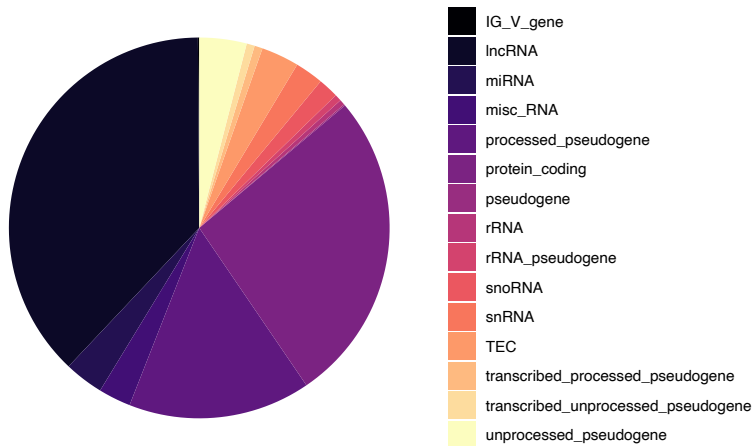
```

大约有 26.61% 的基因是 protein\_coding 类型（计 1 分）。

#### 练习 4

作图比较 21 号染色体上不同类型基因的比例（计 1 分）

Gene Types Proportions



观察结果可以发现，IG\_V\_gene 占比最高，而与核糖体 rRNA 及与其加工修饰相关的 snRNA 和 snoRNA 占比较低。编码蛋白的基因占比略高于 25%，与先前统计结果一致。

选择 3 种你感兴趣的基因类型，解释其功能，需包含参考文献（计 0.5 分）

这里选择 lncRNA、snRNA 以及 miRNA 解释功能。

##### 1. lncRNA

长链非编码 RNA（lncRNA）指对应基因不编码蛋白质的长的 RNA，包括 RNA 聚合酶 I

(Pol I)、Pol II 和 Pol III 转录的 RNA，以及来自加工后的内含子的 RNA。大多数 lncRNA 具有细胞类型特异性，参与调节细胞分化以及其他生理过程。lncRNA 在细胞质中也发挥着重要作用，包括参与翻译、代谢和信号转导的调节。lncRNA 通常具有模块化结构，并且具有丰富的重复，研究逐渐表明这些重复与其功能相关。<sup>[1]</sup>

## 2. snoRNA

小核仁 RNA (snoRNA) 是一种非编码 RNA，广泛存在于真核细胞的核小体中，在 rRNA 修饰中发挥着重要作用，snoRNA 同时也参与 tRNA 和 mRNA 修饰。<sup>[2]</sup>

## 3. miRNA

miRNA 是长度约 22 个核苷酸的 RNA，其来自较长的初级 miRNA (pri-miRNA) 转录物，一般带有一个或多个发夹结构<sup>[3]</sup>。在动物体内，miRNA 的主要通过结合 mRNA 抑制其翻译，从而调控基因的表达。而在植物体内，miRNA 与 siRNA 的作用机制类似，通过结合并剪切 mRNA 使其降解，从而抑制其翻译过程。

## 参考文献

- [1] Mattick, J.S., Amaral, P.P., Carninci, P. *et al.* Long non-coding RNAs: definitions, functions, challenges and recommendations. *Nat Rev Mol Cell Biol* **24**, 430–447 (2023).
- [2] Huang, Zh., Du, Yp., Wen, Jt. *et al.* snoRNAs: functions and mechanisms in biological processes, and roles in tumor pathophysiology. *Cell Death Discov.* **8**, 259 (2022).
- [3] Shang, R., Lee, S., Senavirathne, G. *et al.* microRNAs in action: biogenesis, function and regulation. *Nat Rev Genet* **24**, 816–833 (2023).